Software Requirements Specification (SRS)

Utility Management System

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1. Introduction

1.1 Purpose

This SRS specifies requirements, architecture, and database models for the web-based Utility Management System (UMS).

1.2 Scope

UMS automates multi-utility (Electricity, Water, Gas) operations. Core features:

- **Auth:** Role-based access (`admin`, `employee`, `customer`).
- **Meters:** Smart meter inventory & reading tracking.
- **Billing:** Automated tariff billing & PDF invoices.
- **Notifications:** Instant SMTP email invoice reminders.
- **Services:** Digital payments, complaints, GIS map & reports.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition & Context
SRS	Software Requirements Specification
UMS	Utility Management System Platform
UI / SPA	User Interface / Single Page Application
REST / API	Representational State Transfer / HTTP API
GIS / ERD	Geographic Information System / Entity Relationship Diagram

1.4 References

- IEEE Std 830-1998 & IEEE Std 29148-2018 Standards.
- Express.js, MySQL 8.0, and Leaflet.js Technical Docs.

1.5 Overview

Document organized into 7 sections: System Overview, Functions, Requirements (FR-01 to FR-41), Interfaces, NFRs, Policies, and DFD/ERD.

2. Overall Description

2.1 Product Perspective

UMS is a full-stack SPA. Backend runs on Node.js/Express.js with MySQL (`utilis_db`) via REST API.

2.2 Product Functions

- User authentication & role session control.
- Customer directory & profile management.
- Smart meter registration & classification (Electricity/Water/Gas).
- Monthly reading logs & delta calculation (`curr - prev`).
- Tariff billing calculation & PDF invoice creation.
- SMTP email dispatch for bill alerts & reminders.
- Digital payments (bKash, Nagad, Card, Bank).
- Support ticketing with JSON comment logs.
- Field maintenance task assignment for staff.
- Leaflet GIS spatial outage map visualization.
- Executive analytics dashboard with Chart.js charts.

2.3 User Classes and Characteristics

User Role	Responsibilities & Permissions
System Administrator (admin)	Full control over customer records, meter setup, billing, GIS map, staff assignment, and financial reports.
Utility Field Staff (employee)	Logs monthly meter readings, updates technical tickets, and checks GIS meter locations.
Utility Customer (customer)	Views/pays utility bills online, submits meter readings, logs complaints, and tracks payment receipts.

2.4 Operating Environment

- **Server:** Node.js v18+, Express.js, Nodemailer.
- **Database:** MySQL Server v8.0+ (`utilis_db`).
- **Client:** Chrome, Firefox, Edge, Safari running SPA.

2.5 Design and Implementation Constraints

- Dual-port runtime: Express REST API port 5000, HTTP frontend port 8080.
- InnoDB foreign key constraints across 9 relational tables.
- Non-negative consumption delta enforcement (`currReading >= prevReading`).

2.6 User Documentation

- Installation & API operational guide in root `README.md`.

2.7 Assumptions and Dependencies

- Users have browser internet access; MySQL DB & SMTP servers active.

3. System Features (Functional Requirements)

3.1 Module 1: User Authentication & Profile Management

- FR-01: User login via valid username/email and password.
- FR-02: `sessionStorage` state with role permissions (`admin`, `employee`, `customer`).
- FR-03: Self-registration for customers with meter details.
- FR-04: User profile edit; FR-05: Secure session logout.

3.2 Module 2: Customer Account Management

- FR-06: Customer directory linking profiles to user logins.
- FR-07: Admin customer search & edits; FR-08: Account activation/deactivation.
- FR-09: Assign and re-link smart meters to customer accounts.

3.3 Module 3: Smart Meter & Reading Tracking

- FR-10: Meter inventory with serial, utility type (Electricity/Water/Gas), and status.
- FR-11: Register smart meters; FR-12: Log monthly reading units.
- FR-13: Display meter status flags (`active`, `faulty`, `inactive`).

3.4 Module 4: Consumption Billing & Tariff Calculation

- FR-14: Filter meters by category; FR-15: Auto-calculate `consumption = curr - prev`.
- FR-16: Compute bill totals using tariff rates; FR-17: Enforce due dates & billing months.
- FR-18: Admin manual/batch billing; FR-19: Set status (`unpaid`, `paid`, `overdue`).

3.5 Module 5: Automated Email Notification & Reminder System

- FR-20: Auto-dispatch invoice emails on bill creation.
- FR-21: Admin `/api/send-bill-reminder` for unpaid bills.
- FR-22: Render HTML templates with bill details; FR-23: Store in-app notifications.

3.6 Module 6: Payment Processing & Financial Accounting

- FR-24: Digital payments (bKash, Nagad, Rocket, Bank, Card).
- FR-25: Validate inputs & record transaction IDs (`transactionId`).
- FR-26: Auto-update status to `paid`; FR-27: Generate digital payment receipts.

3.7 Module 7: Complaint Management & Support Ticketing

- FR-28: Customer complaint submission with category selection.
- FR-29: Track ticket status (`Pending`, `In Progress`, `Resolved`); FR-30: Staff assignment.
- FR-31: Interactive JSON comment threads between customer and staff.

3.8 Module 8: Field Maintenance & Technical Operations

- FR-32: Log maintenance requests (Meter Repair, Leak Check, Line Maintenance).
- FR-33: Record maintenance type, customer link, technician ID, and timestamp.
- FR-34: Field staff progress updates; FR-35: Admin dashboard maintenance counters.

3.9 Module 9: Geographic Information System (GIS) & Spatial Mapping

- FR-36: Integrate Leaflet GIS map displaying spatial meter distribution.
- FR-37: Plot custom map markers for meter health & utility categories.
- FR-38: Highlight active utility outage zones and transformer substation nodes.

3.10 Module 10: Reports & Executive Analytics Engine

- FR-39: Render Chart.js analytics for monthly consumption and revenue.
- FR-40: Executive KPIs for customer count, meters, revenue, and unpaid balances.
- FR-41: Export customer ledgers and billing summaries to PDF format.

4. External Interface Requirements

4.1 User Interfaces

Web UI built as responsive SPA using HTML5, CSS3, and JavaScript ES6:

- Role-specific sidebar navigation; KPI summary metric cards.
- Interactive data tables with search & filter; Styled modal dialogs.
- Embedded Leaflet GIS map canvas with zoom & tooltip overlays.

4.2 Hardware Interfaces

- **Server:** Server: 2-Core CPU, 4 GB RAM, 20 GB SSD.
- **Client:** Client: PC, laptop, tablet, or smartphone (1024x768 display).

4.3 Software Interfaces

- OS: Windows 10/11, Linux (Ubuntu 20.04+), or macOS.
- Runtime: Node.js v18.0+, Express.js HTTP framework; DB: MySQL 8.0+ `InnoDB` UTF-8.

4.4 Communication Interfaces

- HTTP/HTTPS REST API exchange (`application/json`) on port 5000; CORS enabled port 8080; SMTP email.

5. Non-Functional Requirements

ID	Category	Specification & Benchmark
NFR-01	Performance	API response < 1.5s for DB queries; < 3.0s for email dispatches.
NFR-02	Security	Passwords hashed securely; REST routes protected with session role checks.
NFR-03	Reliability	InnoDB relational foreign key constraints ensure 99.9% data integrity.
NFR-04	Usability	Responsive layout with clear status badges, toast feedback, and modal forms.
NFR-05	Portability	Cross-browser execution across Chrome, Firefox, Edge, and Safari.

6. Other Requirements

6.1 Regulatory Compliance & Tariff Policies

The system enforces standardized tariff structures approved by national utility commissions, applying per-unit charges based on utility category (Electricity per kWh, Water per Litre, Gas per Unit).

6.2 Data Privacy & Billing Audit Trail

All financial transactions, bill generations, and payments record exact creation timestamps and user links to maintain an immutable accounting audit trail.

7. Appendices

7.1 Appendix A: Data Flow Diagram (DFD Level 1)

- Customer → [Login / Register] → User Authentication → Stores Session → `users` & `customers` DB.
- Field Staff → [Meter Reading Data] → Reading Log Process → Stores Consumption Delta → `meter_readings` DB.
- Billing Engine → [Fetch Readings & Tariff] → Billing Process → Computes Amount → `bills` DB → Sends Email.
- Customer → [Payment Details] → Payment Gateway → Updates Status → `payments` & `bills` DB.
- Customer → [Complaint / Request] → Support Ticket Process → Assigns Staff → `complaints` & `maintenance` DB.
- Administrator → [Request GIS & Analytics] → Reporting Engine → Aggregates Map & Financial Graphs.

7.2 Appendix B: Entity Relationship Diagram (ERD) & Database Schema

The `utilis_db` relational database comprises nine core tables structured as follows:

Table	Keys & Attributes	Functional Purpose
users	id (PK), name, username (UQ), email, password, role, status	Stores user authentication credentials & roles (admin, employee, customer).
customers	customer_id (PK), user_id (FK -> users.id), phone, address	Stores customer profile details linked to login account.
meters	meter_id (PK), customer_id (FK), meter_number (UQ), utilityType, status	Stores smart meter inventory (Electricity, Water, Gas) & customer link.
meter_readings	reading_id (PK), meter_id (FK), month, unit_consumed	Tracks monthly consumption units logged per meter.
bills	bill_id (PK), customer_id (FK), amount, due_date, status, meterId, prevReading, currReading, tariffRate	Stores calculated utility bills, reading deltas, tariff rates, due dates, & status.
payments	payment_id (PK), bill_id (FK), amount, payment_date, method, transactionId	Records payment transactions, channels (bKash, Nagad, Bank), & transaction IDs.
complaints	complaint_id (PK), customer_id (FK), subject, description, status, assignedTo, comments (JSON)	Manages customer support tickets, status pipeline, staff assignment, & comments.
notifications	id (PK), userId (FK -> users.id), message, type, date, isRead	Stores in-app system notifications and read/unread indicators.
maintenance	maintenance_id (PK), customer_id (FK), request_type, description, status, assigned_to	Manages technical field maintenance requests, assigned technicians, & progress.